

MATH 100:

Linear Algebra

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Topics Covered

- ▶ Vector Equations and Matrix Equations
- ▶ $Ax=b$ Equation
- ▶ Solution Sets of Linear Systems

1.3 Vector Equations

Recap

- Augmented matrices lead us to consider rectangular arrays of numbers—matrices.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Definition of matrix

Definition: A **matrix** is a rectangular array of numbers.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

We say that A has m rows and n columns.

The **dimension** (or **size**) of A is $m \times n$ (m -by- n).

The element a_{ij} is called the (i, j) th entry.

Special Matrix Shapes

An $m \times n$ matrix is a **square matrix of order n** if $m = n$.

A $1 \times n$ matrix is a **row matrix** (or **row vector**):

$$(a_1 \quad a_2 \quad \cdots \quad a_n) .$$

A $m \times 1$ matrix is a **column matrix** (or column vector):

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix}.$$

Abbreviation: In **this** course we will often mention vectors. It is a column vector.

Vectors in $\mathbb{R}^2$

A vector with two entries has the form

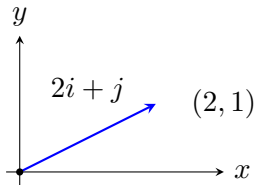
$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad u_1, u_2 \in \mathbb{R}.$$

ordered 2-tuples

The set of all vectors with two entries is denoted by $\mathbb{R}^2$ (read “R-two”).

The $\mathbb{R}$ means the entries are real numbers, and the exponent ² indicates that each vector contains two entries.

Geometric Description of $\mathbb{R}^2$



point $\longleftrightarrow$ vector $\longleftrightarrow$ column matrix

$$(2,1) \qquad 2\mathbf{i} + \mathbf{j} \qquad \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Vector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is the point (x_1, x_2) in the plane.

$\mathbb{R}^2$ is the set of all points in the plane.

Vectors in $\mathbb{R}^3$ and $\mathbb{R}^n$

Vectors in $\mathbb{R}^3$ (vectors with 3 entries)

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (\text{an ordered 3-tuple}).$$

Vectors in $\mathbb{R}^n$ (vectors with n entries)

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad (\text{an ordered } n\text{-tuple}).$$

written as $\mathbf{u}$ or $\vec{u}$.

The Zero Vector $\mathbf{0}$

The zero vector $\mathbf{0}$

The **zero vector** $\mathbf{0}$

The vector whose entries are all zero. $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

The Zero Vector $\mathbf{0}$

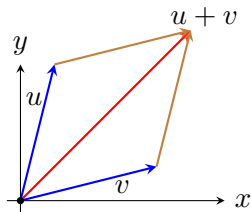
The zero vector $\mathbf{0}$

The vector whose entries are all zero. $\mathbf{0} =$

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Well... appears useless right?

Vector Arithmetic: Addition



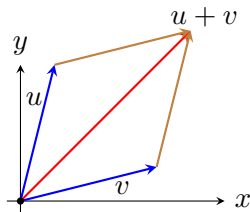
We can view each column as a vector.

$$u = \begin{bmatrix} 0.5 \\ 2 \end{bmatrix}$$

$$v = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

$$u + v = \begin{bmatrix} 2 + 0.5 \\ 2 + 0.5 \end{bmatrix}$$

Vector Arithmetic: Addition



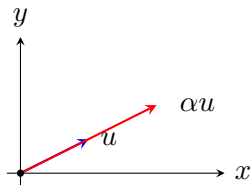
Parallelogram Rule

If $\mathbf{u}$ and $\mathbf{v}$ are points in the $\mathbb{R}^2$ plane:

We have three points represented by $\mathbf{v}$, $\mathbf{u}$, $\mathbf{0}$ (origin).

Three points define a Parallelogram: it ends at $\mathbf{u} + \mathbf{v}$.

Vector Arithmetic: Scalar Multiplication

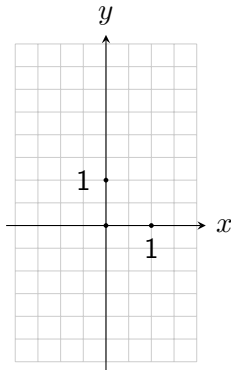


Scalar Multiplication

For a scalar α (this is a number), the vector $\alpha \mathbf{u}$ stretches or shrinks $\mathbf{u}$ correspondingly. Its direction is along the same line, but its length is scaled by $|\alpha|$. If $\alpha < 0$, its direction is reversed.

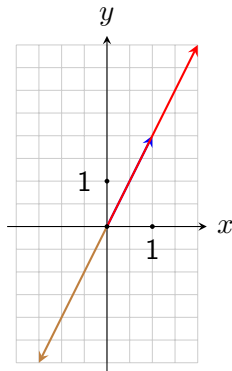
Vector Arithmetic: Scalar Multiplication

Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Express $\mathbf{u}$, $2\mathbf{u}$, and $-\frac{3}{2}\mathbf{u}$ on a graph.



Vector Arithmetic: Scalar Multiplication

Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Express $\mathbf{u}$, $2\mathbf{u}$, and $-\frac{3}{2}\mathbf{u}$ on a graph.



Vector Addition: recap

$$u = \begin{bmatrix} 0.5 \\ 2 \end{bmatrix}$$

$$v = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

$$u + v = \begin{bmatrix} 2 + 0.5 \\ 2 + 0.5 \end{bmatrix}$$

Vector Addition: Geometry and Algebra

Arrows $\longleftrightarrow$ Column Matrices
 Operations on vectors $\longleftrightarrow$ Matrix operations

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}.$$

Scalar Multiplication: recap

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Express $\mathbf{u}$, $2\mathbf{u}$, and $-\frac{3}{2}\mathbf{u}$ on a graph:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad 2\mathbf{u} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad -\frac{3}{2}\mathbf{u} = \begin{bmatrix} -\frac{3}{2} \\ -3 \end{bmatrix}.$$

Scalar Multiplication: Geometry and Algebra

Arrows	$\longleftrightarrow$	Column Matrices
Operations on vectors	$\longleftrightarrow$	Matrix operations
Geometry	$\dashrightarrow$	Algebra

$$\alpha \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} \alpha u_1 \\ \alpha u_2 \end{pmatrix}.$$

Vector Operations in $\mathbb{R}^2$

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$.

- ▶ $\mathbf{u} = \mathbf{v}$ if and only if their corresponding entries are equal. (exactly equal!)
- ▶ $\mathbf{u} + \mathbf{v}$ is obtained by adding corresponding entries.
- ▶ $c\mathbf{u}$, where c is a scalar, is obtained by multiplying each entry in $\mathbf{u}$ by c .

Linear system is defined by solution sets.

Equality, vector addition, and scalar multiplication in $\mathbb{R}^n$ are defined entry by entry, just as in $\mathbb{R}^2$.

Eight Algebraic Properties

For all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and scalars α, β :

$$\text{A1. } \mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$$

$$\text{A2. } (\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$$

$$\text{A3. } \mathbf{x} + \mathbf{0} = \mathbf{x}$$

$$\text{A4. } \mathbf{x} + (-\mathbf{x}) = \mathbf{0}$$

$$\text{A5. } \alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$$

$$\text{A6. } (\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$$

$$\text{A7. } (\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$$

$$\text{A8. } 1\mathbf{x} = \mathbf{x}$$

Linear Combinations

Given $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^n$ and scalars $c_1, \dots, c_p$, the vector

$$\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

is called a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_p$, using weights $c_1, \dots, c_p$.

Examples include

$$\sqrt{3}\mathbf{v}_1 + \mathbf{v}_2, \quad \frac{1}{2}\mathbf{v}_1 = \frac{1}{2}\mathbf{v}_1 + 0\mathbf{v}_2, \quad \mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2.$$

The weights may be any real numbers, including zero.

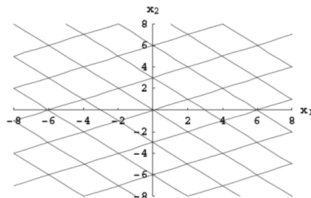
Example: Linear Combinations in $\mathbb{R}^2$

Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}.$$

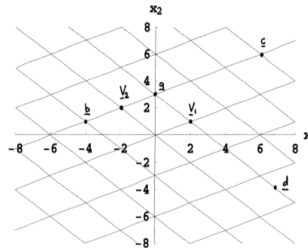
Express each vector as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$:

$$\mathbf{a} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 7 \\ -4 \end{bmatrix}.$$



Example: Solutions

$$\begin{aligned} \mathbf{a} &= \mathbf{v}_1 + \mathbf{v}_2, & \mathbf{b} &= -\mathbf{v}_1 + \mathbf{v}_2, \\ \mathbf{c} &= 4\mathbf{v}_1 + \mathbf{v}_2, & \mathbf{d} &= \mathbf{v}_1 - \frac{5}{2}\mathbf{v}_2. \end{aligned}$$



Example: A Vector Equation

Let

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

Determine whether $\mathbf{b}$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$.

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Determine whether $\mathbf{b}$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$.

This is equivalent to finding weights x_1, x_2, x_3 such that

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + x_3 \mathbf{a}_3 = \mathbf{b}.$$

Rewriting the Vector Equation

Use the definitions of scalar multiplication and vector addition to rewrite the vector equation:

$$x_1 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

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Which is the same as

$$\begin{bmatrix} x_1 \\ 0 \\ 3x_1 \end{bmatrix} + \begin{bmatrix} 4x_2 \\ 2x_2 \\ 14x_2 \end{bmatrix} + \begin{bmatrix} 3x_3 \\ 6x_3 \\ 10x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

From a Vector Equation to a Linear System

$$\begin{bmatrix} x_1 + 4x_2 + 3x_3 \\ 2x_2 + 6x_3 \\ 3x_1 + 14x_2 + 10x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

By equality of vectors,

$$\begin{cases} x_1 + 4x_2 + 3x_3 = -1, \\ 2x_2 + 6x_3 = 8, \\ 3x_1 + 14x_2 + 10x_3 = -5. \end{cases}$$

Solving the Vector Equation

The corresponding **augmented matrix**:

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right]$$

Solving the Vector Equation

The corresponding **augmented matrix**:

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array} \right].$$

Solving the Vector Equation

The corresponding **augmented matrix**:

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array} \right].$$

Thus $x_1 = 1$, $x_2 = -2$, and $x_3 = 2$. Therefore $\mathbf{b}$ is a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$.

$$\mathbf{b} = \mathbf{a}_1 - 2\mathbf{a}_2 + 2\mathbf{a}_3.$$

Vector Equations and Augmented Matrices

A solution of

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{b}$$

is found by solving the linear system whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{b} \end{bmatrix}.$$

Review of the last example: $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ and $\mathbf{b}$ are columns of the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right].$$

$$\begin{array}{cccc} \uparrow & \uparrow & \uparrow & \uparrow \\ \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{b} \end{array}$$

Vector Equation and Linear System

The vector equation

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix}.$$

In particular, $\mathbf{b}$ can be generated by a linear combination of $\mathbf{a}_1, \dots, \mathbf{a}_n$ if and only if the corresponding linear system is consistent.

Vector Equations and Ordinary Systems

Vector equations $\iff$ Ordinary systems of equations

The key idea of linear algebra is to find a solution

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The solution here is to find all **vectors** satisfying the system...

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Vector equations $\iff$ Ordinary systems of equations

The key idea of linear algebra is to find a solution

The solution here is to find all **vectors** satisfying the system...

... that are written as a linear combination of a fixed set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of vectors.

Span $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$

If $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^n$, then the set of all their linear combinations is denoted by

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}.$$

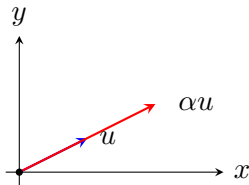
It is the subset of $\mathbb{R}^n$ **spanned** (or **generated**) by $\mathbf{v}_1, \dots, \mathbf{v}_p$; that is, the collection of all vectors of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p,$$

$c_1, \dots, c_p$ being scalars.

Span

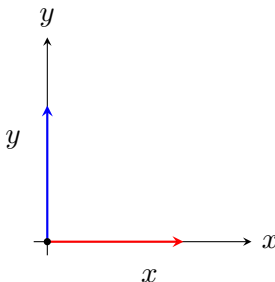
Span is not "a" vector. It tells us what shapes of vectors can be generated:



This only has one direction.

Span

How to use the minimal amount of vectors to span the entire plane?



Combining $\mathbf{x}$ and $\mathbf{y}$ we can reach any point in the plane.

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2.$$

A Geometric Description of $\text{Span}\{\mathbf{v}\}$

Let $\mathbf{v}$ be a nonzero vector in $\mathbb{R}^3$. Then $\text{Span}\{\mathbf{v}\}$ is the set of all scalar multiples of $\mathbf{v}$, namely the line in $\mathbb{R}^3$ through $\mathbf{v}$ and $\mathbf{0}$.

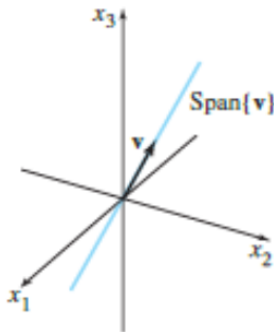
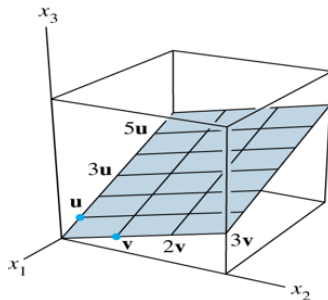


FIGURE 10 $\text{Span}\{\mathbf{v}\}$ as a line through the origin.

A Geometric Description of $\text{Span}\{\mathbf{u}, \mathbf{v}\}$

- ▶ If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in $\mathbb{R}^3$, with $\mathbf{v}$ **not a scalar multiple** of $\mathbf{u}$, then $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is the plane in $\mathbb{R}^3$ containing $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{0}$.
- ▶ In particular, $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ contains the line $\mathbb{R}\mathbf{u}$ through $\mathbf{u}$ and $\mathbf{0}$ and the line $\mathbb{R}\mathbf{v}$ through $\mathbf{v}$ and $\mathbf{0}$.



Example: Is $\mathbf{b}$ in the Span?

Let

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}.$$

Is $\mathbf{b}$ in $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$?

$$\left[\begin{array}{cc|c} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & -18 & 10 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{array} \right].$$

The third equation is $0 = -2$, so the system has no solution. Hence $\mathbf{b} \notin \text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$.

The Matrix Equation

1.4 The Matrix Equation $A\mathbf{x}=\mathbf{b}$

$$A\mathbf{x} = \mathbf{b}$$

Two Ways to Denote an $m \times n$ Matrix

In terms of the entries of A :

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}.$$

In terms of its columns:

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n], \quad \mathbf{a}_i \in \mathbb{R}^m \quad (i = 1, \dots, n).$$

Two Ways to Denote a matrix

In terms of its columns:

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n], \quad \mathbf{a}_i \in \mathbb{R}^m \quad (i = 1, \dots, n).$$

For example:

$$\mathbf{a}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{i1} \\ \vdots \\ a_{m1} \end{bmatrix}.$$

Definition of the Matrix–Vector Product

If A is an $m \times n$ matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and $\mathbf{x} \in \mathbb{R}^n$, then

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Thus $A\mathbf{x}$ is the **linear combination** of the columns of A using the corresponding entries of $\mathbf{x}$ as weights.

$A\mathbf{x}$ is defined only when the number of columns of A equals the number of entries of $\mathbf{x}$. ("Length" of A = "Height" of $\mathbf{x}$)

Moreover, $\mathbf{a}_i \in \mathbb{R}^m$ and $A\mathbf{x} \in \mathbb{R}^m$.

Example: Computing Ax by Columns

$$\underbrace{\begin{bmatrix} 1 & -4 \\ 3 & 2 \\ 0 & 5 \end{bmatrix}}_A \underbrace{\begin{bmatrix} 7 \\ -6 \end{bmatrix}}_x = 7 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} - 6 \begin{bmatrix} -4 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 7 \\ 21 \\ 0 \end{bmatrix} + \begin{bmatrix} 24 \\ -12 \\ -30 \end{bmatrix} = \underbrace{\begin{bmatrix} 31 \\ 9 \\ -30 \end{bmatrix}}_b.$$

Here, A

Example: Write a Linear Combination as $A\mathbf{x}$

For $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^m$, write the linear combination $3\mathbf{v}_1 - 5\mathbf{v}_2 + 7\mathbf{v}_3$ as a matrix times a vector.

Example: Write a Linear Combination as $A\mathbf{x}$

For $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^m$, write the linear combination $3\mathbf{v}_1 - 5\mathbf{v}_2 + 7\mathbf{v}_3$ as a matrix times a vector.

Answer: place $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ into the columns of A and the weights $3, -5, 7$ into $\mathbf{x}$:

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Answer: place $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ into the columns of A and the weights $3, -5, 7$ into $\mathbf{x}$:

$$3\mathbf{v}_1 - 5\mathbf{v}_2 + 7\mathbf{v}_3 = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix} = A\mathbf{x}.$$

Three Forms of the Same Linear System

$$\begin{cases} 2x_1 + 3x_2 + 4x_3 = 9, \\ -3x_1 + x_2 = -2. \end{cases}$$

Vector equation:

$$x_1 \begin{bmatrix} 2 \\ -3 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 \\ -2 \end{bmatrix}.$$

Matrix equation:

$$\begin{bmatrix} 2 & 3 & 4 \\ -3 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -2 \end{bmatrix}, \quad A\mathbf{x} = \mathbf{b}.$$

Three Equivalent Views of a Linear System

A linear system may be viewed

- ① as a system of linear equations;
- ② as a vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b};$$

- ③ as a matrix equation $A\mathbf{x} = \mathbf{b}$.

Theorem 3

If A is an $m \times n$ matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and $\mathbf{b} \in \mathbb{R}^m$, then the matrix equation

$$A\mathbf{x} = \mathbf{b}$$

has the same solution set as the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b},$$

which has the same solution set as the linear system with augmented matrix

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix}.$$

Consequently, $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is a linear combination of the columns of A .

Theorem 4 and Example

For an $m \times n$ matrix A , the following are logically equivalent:

- ① For every $\mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
- ② Every $\mathbf{b} \in \mathbb{R}^m$ is a linear combination of the columns of A .

How to determine an equation is consistent for all vectors

EXAMPLE: Let

$$A = \begin{bmatrix} 1 & 4 & 5 \\ -3 & -11 & -14 \\ 2 & 8 & 10 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all $\mathbf{b}$?

Answer: augmented matrix corresponding to $A\mathbf{x} = \mathbf{b}$:

$$\left[\begin{array}{ccc|c} 1 & 4 & 5 & b_1 \\ -3 & -11 & -14 & b_2 \\ 2 & 8 & 10 & b_3 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 4 & 5 & b_1 \\ 0 & 1 & 1 & 3b_1 + b_2 \\ 0 & 0 & 0 & -2b_1 + b_3 \end{array} \right].$$

(Can you find a value of b_1 and b_3 that makes $-2b_1 + b_3$ non zero?)

Hence $A\mathbf{x} = \mathbf{b}$ is **not** consistent for every $\mathbf{b}$.

Exercise: Consistency for Every $\mathbf{b}$?

Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Is $A\mathbf{x} = \mathbf{b}$ consistent for every possible $\mathbf{b}$?

$$\left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 3 & 4 & b_2 \\ 5 & 6 & b_3 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & -2b_1 + b_2 \\ 0 & 1 & \frac{3b_1 - b_2}{2} \\ 0 & 0 & b_1 - 2b_2 + b_3 \end{array} \right].$$

The equation is not consistent for every $\mathbf{b}$ because some choices make $b_1 - 2b_2 + b_3 \neq 0$.

Properties of the Matrix–Vector Product $A\mathbf{x}$

Theorem 5

If A is an $m \times n$ matrix, $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, and c is a scalar, then

- ① $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$;
- ② $A(c\mathbf{u}) = cA\mathbf{u}$.

Computation of $A\mathbf{x}$

Let

$$A = \begin{pmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

From the definition,

$$\begin{aligned} A\mathbf{x} &= x_1 \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix} + x_2 \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} + x_3 \begin{pmatrix} 4 \\ -3 \\ 8 \end{pmatrix} \\ &= \begin{pmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{pmatrix}. \end{aligned}$$

Computing $A\mathbf{x}$ Row by Row

$$\begin{pmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{pmatrix}.$$

Each output entry is the dot product of one row of A with $\mathbf{x}$:

$$(2, 3, 4)\mathbf{x} = 2x_1 + 3x_2 + 4x_3,$$

$$(-1, 5, -3)\mathbf{x} = -x_1 + 5x_2 - 3x_3,$$

$$(6, -2, 8)\mathbf{x} = 6x_1 - 2x_2 + 8x_3.$$

Row-Vector Rule for Computing $A\mathbf{x}$

If $A\mathbf{x}$ is defined, then its i th entry is the sum of the products of corresponding entries from row i of A and from $\mathbf{x}$.

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix},$$

$$\begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ s \\ t \end{bmatrix} = \begin{bmatrix} r \\ s \\ t \end{bmatrix}.$$

Identity Matrix

The $n \times n$ identity matrix is

$$I = I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Its diagonal entries are 1 and all other entries are 0. For example,

$$I_3 \begin{bmatrix} r \\ s \\ t \end{bmatrix} = \begin{bmatrix} r \\ s \\ t \end{bmatrix}.$$

In general,

$$I_n \mathbf{x} = \mathbf{x} \quad \text{for every } \mathbf{x} \in \mathbb{R}^n.$$

Solution Sets of Linear Systems

1.5 Solution Sets of Linear Systems

Topics

- ▶ Parametric Vector Form
- ▶ Solution Sets of Homogeneous Linear Systems
- ▶ Solution Sets of Nonhomogeneous Systems

Parametric Vector Form

A parametric description

$$\begin{cases} x_1 = 13x_3 - 5x_4, \\ x_2 = 8x_3 - 3x_4, \\ x_3 \text{ is free,} \\ x_4 \text{ is free} \end{cases}$$

becomes, on setting $x_3 = c_1$ and $x_4 = c_2$,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = c_1 \underbrace{\begin{bmatrix} 13 \\ 8 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{u}} + c_2 \underbrace{\begin{bmatrix} -5 \\ -3 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}}.$$

Two Descriptions of a Solution Set

Parametric description

A parametric description defines the coordinates of a curve, surface, or system using independent variables called parameters rather than a direct relationship between coordinates

Parametric vector description

An explicit description of the plane as the set spanned by $\mathbf{u}$ and $\mathbf{v}$.

Solution Sets of Homogeneous Linear Systems

A **homogeneous system** has the form

$$A\mathbf{x} = \mathbf{0},$$

where A is $m \times n$ and $\mathbf{0}$ is the zero vector in $\mathbb{R}^m$.

Example:

$$\begin{cases} x_1 + 10x_2 = 0, \\ 2x_1 + 20x_2 = 0, \end{cases} \quad \begin{bmatrix} 1 & 10 \\ 2 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Trivial and Nontrivial Solutions

The homogeneous system $A\mathbf{x} = \mathbf{0}$ always has the **trivial solution** (Trivial: simple, obvious, expected answer).

$$\mathbf{x} = \mathbf{0}.$$

Here $\mathbf{x}$ and $\mathbf{0}$ are both vectors. Basically, it is setting all variables to 0. It always works! Nonzero vector solutions are called **nontrivial solutions**.

For the previous example,

$$\left[\begin{array}{cc|c} 1 & 10 & 0 \\ 2 & 20 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 10 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

The system is consistent and has a free variable, so it has infinitely many solutions, including nontrivial ones.

When Do Nontrivial Solutions Exist?

A homogeneous equation $A\mathbf{x} = \mathbf{0}$ has nontrivial solutions if and only if the system has **at least one free variable**.

Determine whether the following system has nontrivial solutions, and then describe its solution set:

$$\begin{cases} 2x_1 + 4x_2 - 6x_3 = 0, \\ 4x_1 + 8x_2 - 10x_3 = 0. \end{cases}$$

There are three variables but at most two pivots, so at least one variable is free; hence nontrivial solutions exist.

When Do Nontrivial Solutions Exist?

Why we need to have free variables for nontrivial solution to exist?

Row Reduction of the Homogeneous System

$$\left[\begin{array}{ccc|c} 2 & 4 & -6 & 0 \\ 4 & 8 & -10 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 4 & 8 & -10 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right].$$

Therefore

$$x_1 = -2x_2, \quad x_2 \text{ is free}, \quad x_3 = 0.$$

Parametric Vector Description

Since $x_1 = -2x_2$, x_2 is free, and $x_3 = 0$,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{v}} = x_2 \mathbf{v}.$$

Thus the solution set is

$$\text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Solution Sets of Nonhomogeneous Linear Systems

Describe the solution set of

$$\begin{cases} 2x_1 + 4x_2 - 6x_3 = 0, \\ 4x_1 + 8x_2 - 10x_3 = 4. \end{cases}$$

Nonhomogeneous because (0,4) is not all zero (0,0).

Row reduction gives

$$\left[\begin{array}{ccc|c} 2 & 4 & -6 & 0 \\ 4 & 8 & -10 & 4 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 6 \\ 0 & 0 & 1 & 2 \end{array} \right].$$

Hence

$$\mathbf{x} = \begin{bmatrix} 6 - 2x_2 \\ x_2 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 6 \\ 0 \\ 2 \end{bmatrix}}_{\mathbf{p}} + x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{v}} = \mathbf{p} + 2\mathbf{v}$$

Recap of the Previous Two Examples

The solution set of $A\mathbf{x} = \mathbf{0}$ is

$$\mathbf{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = x_2 \mathbf{v}.$$

The solution set of $A\mathbf{x} = \mathbf{b}$ is

$$\mathbf{x} = \begin{bmatrix} 6 \\ 0 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \mathbf{p} + x_2 \mathbf{v}.$$

Theorem 6

Suppose $A\mathbf{x} = \mathbf{b}$ is consistent (exists solution) for some given $\mathbf{b}$ and let $\mathbf{p}$ be one particular solution. Then its solution set is the set of all vectors of the form

$$\mathbf{w} = \mathbf{p} + \mathbf{v}_h,$$

where $\mathbf{v}_h$ is any solution of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Warning: This theorem applies only when $A\mathbf{x} = \mathbf{b}$ has at least one solution. If the $A\mathbf{x} = \mathbf{b}$ is inconsistent, its solution set is empty.

CHECK IF SOLUTION EXISTS FIRST!

Example: A Homogeneous Plane

Describe the solution set of $2x_1 - 4x_2 - 4x_3 = 0$ and compare it with the solution set of $2x_1 - 4x_2 - 4x_3 = 6$.

Example: A Homogeneous Plane

Describe the solution set of $2x_1 - 4x_2 - 4x_3 = 0$ and compare it with the solution set of $2x_1 - 4x_2 - 4x_3 = 6$.

Solution: for the first homogeneous equation,

$$\begin{bmatrix} 2 & -4 & -4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 & 0 \end{bmatrix}.$$

Therefore

$$\mathbf{x} = \begin{bmatrix} 2x_2 + 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{u}} + x_3 \underbrace{\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}}.$$

Example: A Parallel Nonhomogeneous Plane

For the second nonhomogeneous equation $2x_1 - 4x_2 - 4x_3 = 6$,

$$\begin{bmatrix} 2 & -4 & -4 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 & 3 \end{bmatrix}.$$

Thus

$$\mathbf{x} = \begin{bmatrix} 3 + 2x_2 + 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{p}} + x_2 \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{u}} + x_3 \underbrace{\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}}.$$

Exercise: True or False?

- ① If $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.

Exercise: True or False?

- ① If $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then $A\mathbf{x} = \mathbf{b}$ has a unique solution. **False**
Has 0 or 1 solution
- ② If $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.

Exercise: True or False?

- ① If $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then $A\mathbf{x} = \mathbf{b}$ has a unique solution. **False**
Has 0 or 1 solution
- ② If $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. **False**
Has 0 or infinitely many
- ③ If $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

Exercise: True or False?

- ① If $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then $A\mathbf{x} = \mathbf{b}$ has a unique solution. **False**
Has 0 or 1 solution
- ② If $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. **False**
Has 0 or infinitely many
- ③ If $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution. **True**
Only trivial

Exercise: True or False?

● If $A\mathbf{x} = \mathbf{b}$ has a unique solution, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

Exercise: True or False?

- If $A\mathbf{x} = \mathbf{b}$ has a unique solution, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution. **False**
Only nontrivial
- Any $n \times n$ system of linear equations has at most n solutions.

Exercise: True or False?

- Ⓐ If $A\mathbf{x} = \mathbf{b}$ has a unique solution, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution. **False**
Only nontrivial
- Ⓑ Any $n \times n$ system of linear equations has at most n solutions. **False**
May be 0, 1, or infinitely many
- Ⓒ The system $A\mathbf{x} = \mathbf{0}$ has the trivial solution if and only if there is no free variable.

Exercise: True or False?

- If $A\mathbf{x} = \mathbf{b}$ has a unique solution, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution. **False**
Only nontrivial
- Any $n \times n$ system of linear equations has at most n solutions. **False**
May be 0, 1, or infinitely many
- The system $A\mathbf{x} = \mathbf{0}$ has the trivial solution if and only if there is no free variable. **False**
Completely reversed. If and only if there is at least one free variable